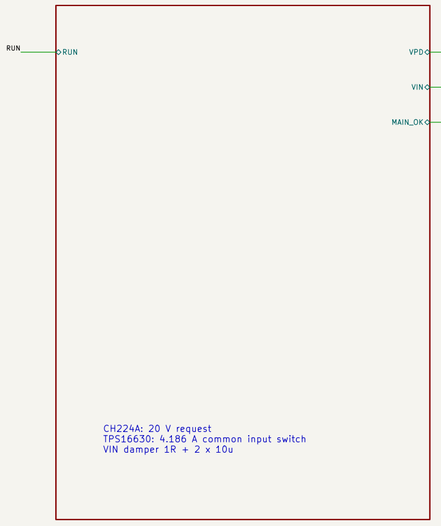
USB POWER 45W S9.0 / SINGLE BOARD, +/−12V ONLY

S9.0: +5V subsystem removed; +/−12V directly from the converters; ceramic banks; two parallel outputs; VPD−powered OV guard (+12V / −12V); one−shot arming latch; MAIN LED only.

USB PD, main eFuse and VIN damper



File: 01_usb_input.kicad_sch

+12 V converter



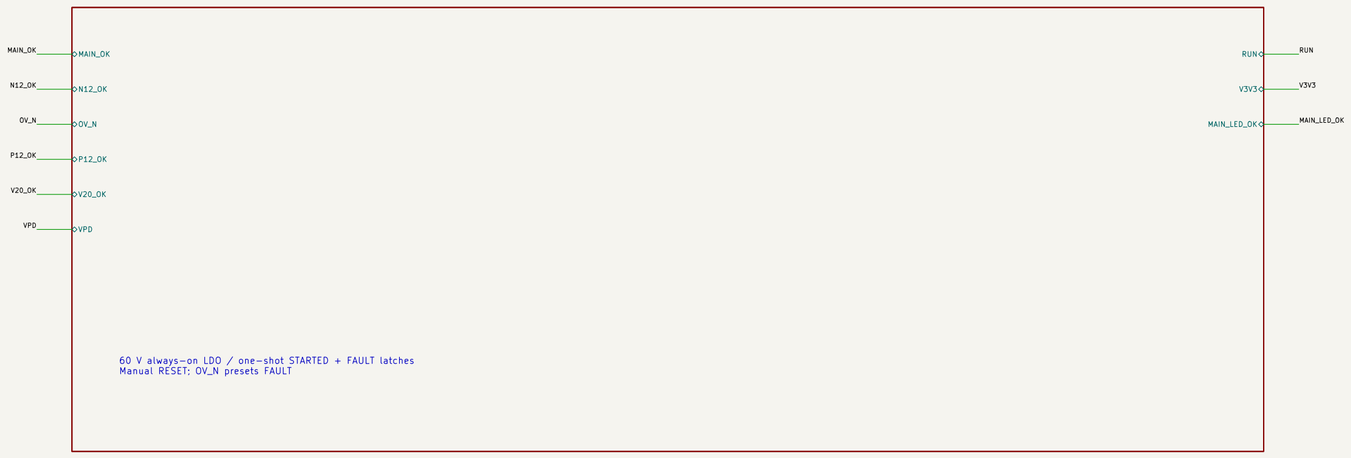
File: 02_positive_12v.kicad_sch

−12 V inverting converter



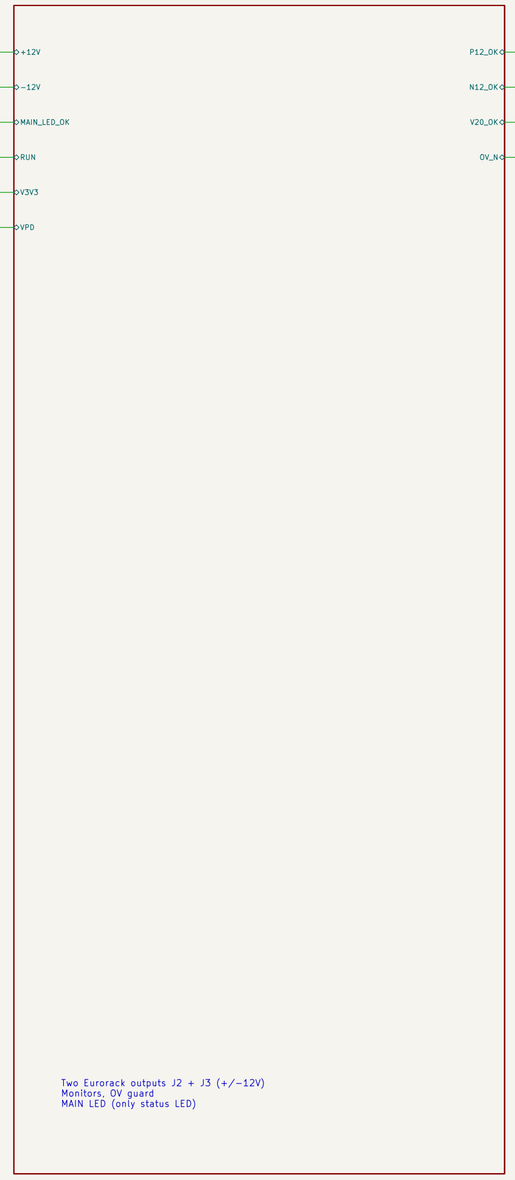
File: 03_negative_converter.kicad_sch

Always−on control − fault latch, overvoltage preset and RESET



File: 04_fault_reset.kicad_sch

Two output headers − monitors, overvoltage guard and MAIN LED



File: 05_outputs_status.kicad_sch

FAULT POLICY
One−shot qualification from the first converter enable (STARTED latch): after
0.31−0.88 s any persistent bad status −> all OFF until RESET. Severe OV via U20 acts even before ARMED.
S9.0: ANY_BAD = NOT(MAIN_LED_OK AND P12_OK AND N12_OK); no +5V rail.
MAIN indicates enabled/status, not precision regulation.

OUTPUTS: +12 / −12 on J2 and J3 in parallel (pins 11/12 open)
Published conservative rating (prototype): see docs/s8−single−board.md;
above 2 A per rail use both headers.

One−shot arming latch (STARTED): U18/U20 on VPD; see docs/s8−single−board.md
S9.0 single board: +/−12V only (+5V subsystem removed); MAIN LED only

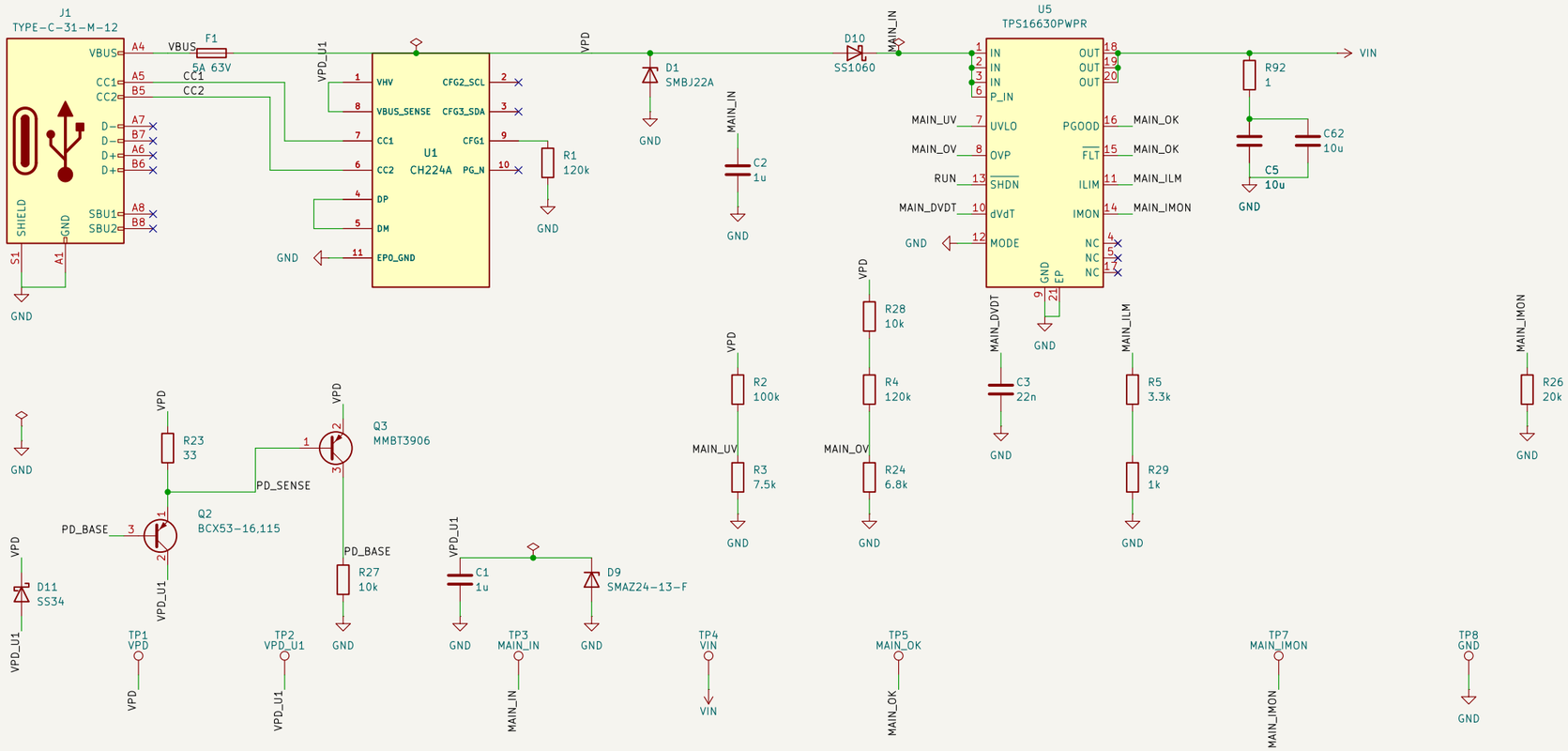
Big Modular

Sheet: /
File: usb−power−45w−single.kicad_sch

Title: USB power S9 single board − wired power flow overview

Size: User	Date:	**Rev: S9.0**
KiCad E.D.A. 10.0.6		Id: 1/6

USB PD, MAIN EFUSE AND VIN DAMPER



20 V PD request: U5 TPS16630 (60 V, 4.186 A nominal) is the common input switch: RUN low turns every rail off.
MAIN_OK = FLT + PGOOD at 3.3 V. R92 1R + C5/C62 2 x 10u damp the all-ceramic VIN bus (replaces the 47u can).
S9.0: the U5 nets are named MAIN_* on the single board (EF_* in S7).

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK MAIN_OK

RUN RUN

VIN VIN

VPD VPD

S9.0: +/-12V only (+5V removed), one-shot arming latch - docs/s8-single-board.md

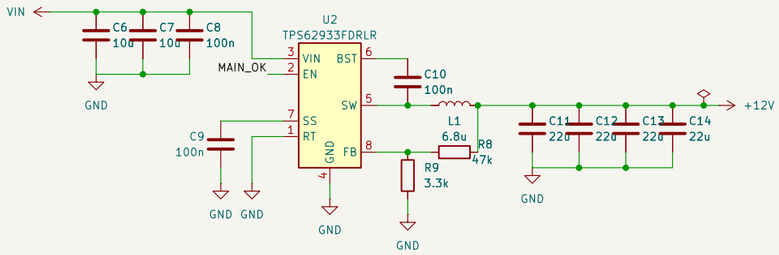
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Sheet: /USB PD, main eFuse and VIN damper/
File: 01_usb_input.kicad_sch

Title: USB PD, main eFuse and VIN damper

Size: User	Date:	**Rev: S9.0**
KiCad E.D.A. 10.0.6		Id: 2/6

+12 V CONVERTER





TPS62933F +12 V, unchanged power stage; its output IS the delivered +12V (no rail eFuse).
Soft start C9 100n = ~14.5 ms into rack capacitance. Own protection: 4.2–5.8 A peak limit, hiccup, OVP 112–118 %, TSD.
Rail ceramics: C11–C14 here + C34 (J2) and C59 (J3) on page 05; the 100u can is gone.

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V ◇—— +12V

MAIN_OK ◇—— MAIN_OK

VIN ◇—— VIN

S9.0: +/−12V only (+5V removed), one-shot arming latch — docs/s8-single-board.md

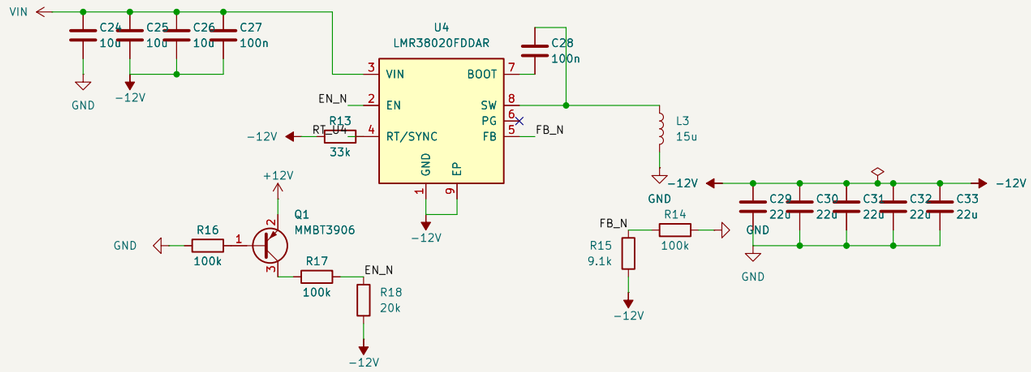
Big Modular

Sheet: /+12 V converter/
File: 02_positive_12v.kicad_sch

Title: +12 V converter

Size: User	Date:	**Rev: S9.0**
KiCad E.D.A. 10.0.6		Id: 3/6

−12 V INVERTING CONVERTER





LMR38020F inverting buck−boost; IC ground / exposed pad = the delivered −12V (no series switch).
4 ms soft start, 18 ms OCP blanking at start−up. Q1 follows +12V. C29−C33 here + C36 (J2) and C61 (J3).
S8.4: hiccup needs |Vout| < 40 % after the blanking. Rack allowance 0.85 mF (TI 1000 uF COUT guidance; hiccup model
at the U5 turn−on minimum, 1 A) covers a typical 0.44 mF rack. Bench start−up test still required.

INTER−SHEET INTERFACE (see wired power−flow overview)

+12V ◇ +12V

−12V ◇ −12V

VIN ◇ VIN

S9.0: +/−12V only (+5V removed), one−shot arming latch − docs/s8−single−board.md

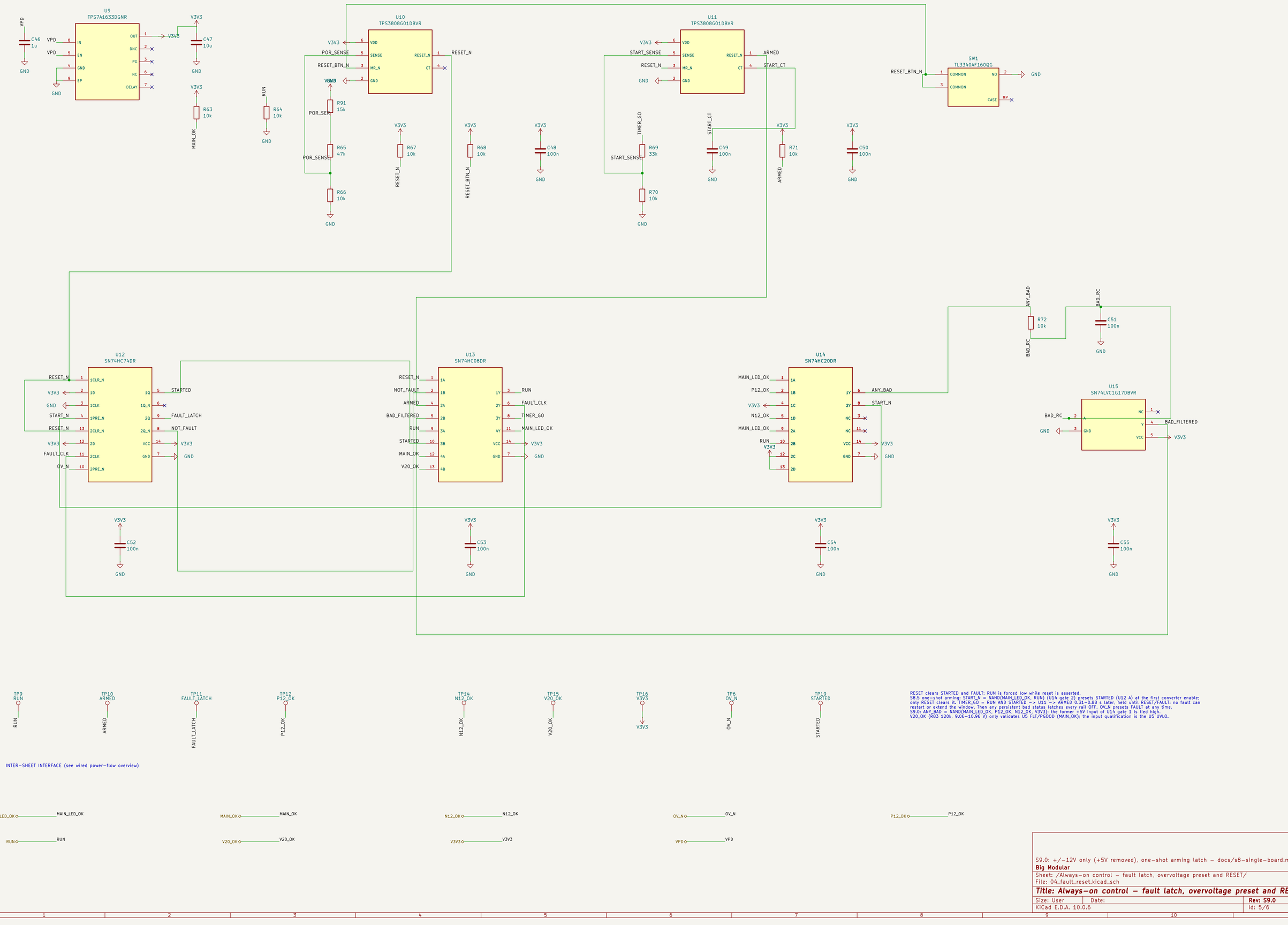
Big Modular

Sheet: /−12 V inverting converter/
File: 03_negative_converter.kicad_sch

Title: −12 V inverting converter

Size: User	Date:	**Rev: S9.0**
KiCad E.D.A. 10.0.6		Id: 4/6

ALWAYS-ON CONTROL — FAULT LATCH, OVERVOLTAGE PRESET AND RESET



TP9 RUN · TP10 ARMED · TP11 FAULT_LATCH · TP12 P12_OK · TP14 N12_OK · TP15 V20_OK · TP16 V3V3 · TP6 OV_N · TP19 STARTED

RESET clears STARTED and FAULT; RUN is forced low while reset is asserted.
S8.5 one-shot arming: START_N = NAND(MAIN_LED_OK, RUN) (U14 gate 2) presets STARTED (U12 A) at the first converter enable; only RESET clears it. TIMER_GO = RUN AND STARTED -> U11 -> ARMED 0.31–0.88 s later, held until RESET/FAULT: no fault can restart or extend the window. Then any persistent bad status latches every rail OFF. OV_N presets FAULT at any time.
S9.0: ANY_BAD = NAND(MAIN_LED_OK, P12_OK, N12_OK, V3V3): the former +5V input of U14 gate 1 is tied high.
V20_OK (RB3 120k, 9.06–10.96 V) only validates U5 FLT/PGOOD (MAIN_OK); the input qualification is the U5 UVLO.

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_LED_OK · MAIN_OK · N12_OK · OV_N · P12_OK · RUN · V20_OK · V3V3 · VPD

S9.0: +/-12V only (+5V removed), one-shot arming latch — docs/s8-single-board.md

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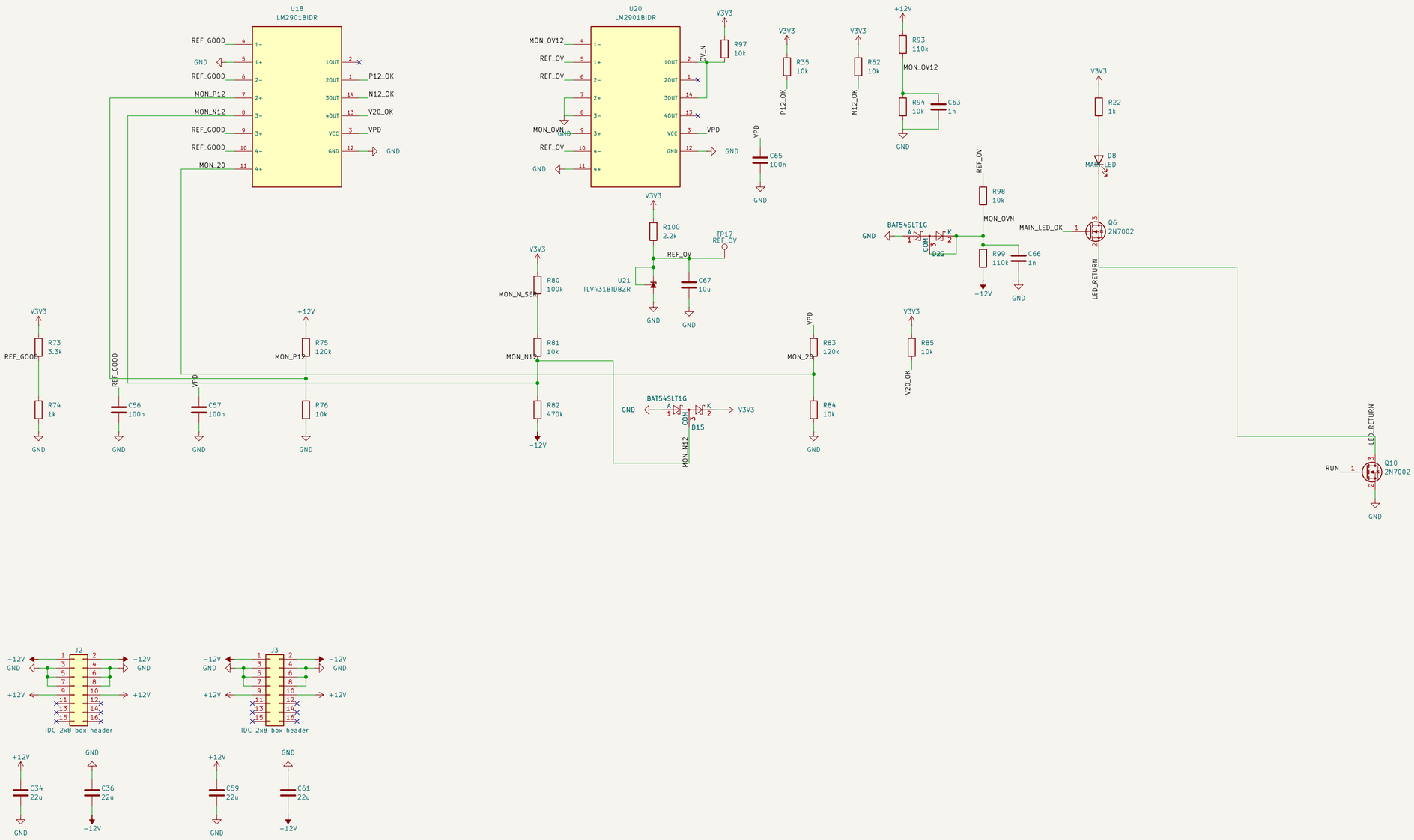
Sheet: /Always-on control — fault latch, overvoltage preset and RESET/
File: 04_fault_reset.kicad_sch

Title: Always-on control — fault latch, overvoltage preset and RESET

Size: User	Date:	**Rev: S9.0**
KiCad E.D.A. 10.0.6		Id: 5/6

TWO OUTPUT HEADERS — MONITORS, OVERVOLTAGE GUARD AND MAIN LED

J2 + J3: two parallel 16-pin outputs, pin 1 = −12V. C34/C36 at J2, C59/C61 at J3 (22u each, replace the 100u cans).
S9.0: +/−12V only; J2/J3 pins 11/12 (+5V) and CV/GATE (13–16) are open.
D8 MAIN is the only status LED (rail LEDs removed): MAIN_LED_OK (U5 on, FLT/PGOOD valid) AND RUN. Any bad status after ARMED or a U20 guard trip latches every rail off. P12_OK / N12_OK: logic + test pads.
U20 overvoltage guard (S8.1): +12V > 14.88 V or −12V < −13.64 V nominal (corners 14.48–15.27 / 13.27–14.02 V) pulls OV_N -> FAULT -> U5 off. Own reference REF_OV = U21 TLV431B (R100, C67, TP17); U18/U20 LM2901B: R93, R94, R98, R99 are 0.1 % / 25 ppm. The −12V channel compares MON_OVN with GND (no V3V3 term); D22 holds MON_OVN near −0.24 V (25 C only, <=0.15 mA) if V3V3 is lost while the rack holds −12V. S9.0: the former +5V comparators of U18/U20 are spares.
S8.3: U18/U20 supplies are always-on VPD; outputs remain 3.3 V logic. No prototype qualification is claimed.



INTER-SHEET INTERFACE (see wired power-flow overview)

+12V — +12V
−12V — −12V
MAIN_LED_OK — MAIN_LED_OK
N12_OK — N12_OK
OV_N — OV_N
P12_OK — P12_OK
RUN — RUN
V20_OK — V20_OK
V3V3 — V3V3
VPD — VPD

S9.0: +/−12V only (+5V removed), one-shot arming latch — docs/s8-single-board.md
Big Modular
Sheet: /Two output headers — monitors, overvoltage guard and MAIN LED/
File: 05_outputs_status.kicad_sch
Title: Two output headers — monitors, overvoltage guard and MAIN LED
Size: User | Date: | **Rev: S9.0**
KiCad E.D.A. 10.0.6 | Id: 6/6